SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1506

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I M.Pranay (192525387) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Pranay

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

CO-1

Assessment Tool-1

Short Answer Test

Name :- M.Pranay

Reg No :- 192525382

Course Code :- CSA1506

Course name :- Cloud computing and Big data Analytics.

1) Define cloud computing and explain any four characteristics that make it different from traditional on Premise computing.

sol:- Cloudcomputing is the on demand delivery of computing services - including servers, storage, databases, networking, and software - over the internet with Pay-as-you-go Pricing.

Four key characteristics Vs. on Premise:-

→ on- Demand self-service :- Users can Provision computing capabilities automatically without requiring human interaction with the service Provider.

Elasticity and scalability :- Resources can be scaled up or down rapidly based on demand: on Premise requires manual hardware upgrades.

Pay-as-you-go Pricing :- You Pay only for what You use (OPEX), where as on-Premise requires significant upfront Capital investment.

Broad Network Access :- services are available over the network and accessed through standard mechanisms.

2) Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled moderan cloud Plattforms

Sol:- Hardware Evolution:-

Large mainframe computers were used in organizations.

Personal computers:-

Affordable PCs made computing accessible to individuals.

client - server Architecture:-

clients ~~retutr~~ requested services from centralized servers.

virtualization:- Multiple virtual machines ran on a single physical server, improving efficiency.

Internet Evolution:-

High - Speed Internet enabled remote access to applications and data.

web Applications:-

Software became availble through web browsers.

cloud computing:-

Providers like AWS, Microsoft Azure, and Google cloud began offering scalable services over the Internet.

3) Explain the role of server virtualization in cloud computing. Differentiate b/w virtualization as an enabling technology and cloud computing as a service model.

Sol:-

Role of server virtualization:-

-) Divides one Physical server into multiple virtual machines.

-) Improves hardware utilization.

-) Reduces infrastructure cost.

-) Enables quick deployment and easy management of resources.

Virtualization	Cloud computing
→ Technology that creates virtual creates virtual machines	→ Services model that delivers computing resources over the Internet.
→ Focuses on efficient use of hardware	→ Focuses on Providir services to users.
→ Runs on a Physical server	→ uses virtualization to Provide Scalable services
Ex:- VMware, Hyper-V	Ex:- Aws, Microsoft. Azure.

4) Discuss the role of data centers in delivering cloud services.

Sol: Data centers are facilities that contain servers, storage devices, networking equipment, and cooling systems used to Provide cloud services.

Roles:-

→ Store and Process user data.
→ Host cloud application and websites
→ Provide secure backup and disaster recovery.
→ Ensure high availability and reliability.
→ Enable virtualization and resource sharing.
→ Deliver cloud services to users world wide through the Internet.

5) Differentiate ~~Iaas~~ Iaas, Paas, SaaS and Xaas with one suitable real-world example for each model.

sol:

Service Model	Description	Example
1) IaaS (Infrastructure as a service)	Provides virtual servers, storage and networking	Amazon EC2 (AWS)
2) Paas (Platform as a service)	Provides a Platform to develop and deploy applications.	Google App Engine
3) SaaS (software as a service)	Delivers software applications over the Internet	Google Docs
4) Xaas (Anything as a service)	Provides various IT services through the cloud	Microsoft 365.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow